Rotation periods and spin senses from Gaia DR3 sparse photometry: tested reliability, 790 candidate new periods and disputed LCDB entries

Oleg Tropynin

Independent researcher, Karaganda, Kazakhstan

Abstract

The error rate of Gaia DR3 sparse-photometry periods for asteroids without a published period has not been measured. I derive periods and spin senses for numbered asteroids with at least 25 Gaia DR3 transits, fitting season offsets, a linear phase term and a second-order Fourier series with the phase corrected for the phase-angle bisector under both senses of rotation. A probability that each period is correct is calibrated on half of the LCDB and tested on the other half, and the confidence threshold is set by null tests that keep the Gaia sampling. 6,238 periods reach the confident tier; 1,680 are absent from the three earlier DR3 analyses. On the test half, 89.3% are correct within 3% (95.4% where the reference quality is $U \geq 3-$); overall, 92.5% agree with independent ATLAS periods. Spin senses agree with independent ATLAS poles for 96.4%, mainly for poles far from the ecliptic; 1,447 are for asteroids with no pole in six pole catalogues. They reproduce the known Yarkovsky segregation in families: 74.8% of 2,345 members lie on the predicted side, against 52.4 ± 0.9% for permuted senses. 39 LCDB periods disagree with independent survey data and 69 more are flagged by agreement among Gaia DR3 analyses. Finally, 790 periods appear in none of the sources searched, including SsODNet. An estimated 4–5% of them are noise and 61–87% are correct; most of the rest are probably harmonics or aliases.

Keywords: Asteroids, Rotation, Photometry, Gaia, Yarkovsky, Lightcurves

Email address: tropininoleg@gmail.com (Oleg Tropynin)

1. Introduction

Gaia (Gaia Collaboration et al., 2016) released sparse photometry for about 157,000 numbered asteroids in DR3 (Gaia Collaboration et al., 2023a,b). Sparse photometry can constrain periods and spin states (Kaasalainen, 2004), and three groups have derived rotation periods and poles from DR3 (Ďurech & Hanuš, 2023; Cellino et al., 2024; MacLennan et al., 2026). Ďurech & Hanuš (2023) estimated a false-solution rate of about 0.6% against LCDB periods of quality $U = 3$. That rate refers to asteroids with a reliable published period. For an asteroid that has no published period, the rate is not known, and it is the rate a user of a new period needs.

This paper sets out to measure it and to deliver the products that depend on it. The reliability of each period is calibrated on one half of the LCDB, tested on the other half, and bounded by null tests that keep the real Gaia sampling. The paper then gives: (i) for 6,238 asteroids, a period with a tested probability of being correct, 1,680 of them absent from all three earlier DR3 analyses; (ii) a spin sense for each of them, 1,447 for asteroids with no pole in six pole catalogues, checked against independent ATLAS poles; (iii) 39 LCDB periods that disagree with independent survey data, and 69 more flagged by agreement among Gaia DR3 analyses; (iv) 790 candidate new periods, with their noise and alias rates. The pipeline runs on all of DR3 in about 20 minutes on a laptop with ten processes, so it can be rerun as soon as Gaia DR4 is released. A pilot on 600 asteroids was reported in Tropynin (2026a).

2. Data and geometry

All 20 bulk files of `sso_observation` (23.3 M CCD rows) were averaged per `transit_id`. This gives 3,069,170 transits of 156,789 numbered asteroids, of which 42,195 have at least 25 transits. Each transit carries the largest relative flux error of its CCDs, a conservative choice that covers CCD-to-CCD scatter, added in quadrature to a 0.03 mag floor. A season is a run of transits with no gap longer than 60 d. Before the search, transits deviating from a fit of season offsets and a phase term by more than 5×1.4826 times the median absolute deviation are removed, and the cut is repeated once; 353 asteroids then fall below 25 transits and are dropped.

Heliocentric positions come from JPL osculating elements (epoch 2026 June), integrated back to 2014–2017 with a fourth-order Runge–Kutta scheme and a one-day step, with the four giant planets as perturbers. Gaia's position

is taken from the data, and light time is iterated. Against JPL Horizons on 19 asteroids, the reduced magnitudes differ by at most 0.0007 mag.

The analysis code was written with the help of a generative AI assistant (Claude, Anthropic) under the author's direction. Its output was checked independently of the code: against JPL Horizons as above, by the held-out TEST half, the null tests, the injections, and comparison with independent surveys. The code is public (see Data availability).

3. Model and search

The reduced magnitude $G - 5\log_{10}(r\Delta)$ is fitted by weighted least squares with one constant per observing season, a term linear in phase angle, and a second-order Fourier series in the phase $2\pi(ft - s\,\lambda_{\rm PAB}/360^\circ)$. Here $\lambda_{\rm PAB}$ is the ecliptic longitude of the phase-angle bisector seen from Gaia (Harris et al., 1984) and $s = \pm 1$ is the sense of rotation. Both senses are searched and the better fit is kept. The correction is exact for a pole at the ecliptic pole, so the periods are approximately sidereal.

Over a season $\lambda_{\rm PAB}$ advances almost linearly, and a linear advance is equivalent to a small shift of frequency, the same for both senses up to sign. The sense is therefore constrained by the non-linear part of $\lambda_{\rm PAB}(t)$ across seasons, and only when the pole is far from the ecliptic; Section 5.3 shows how the accuracy depends on pole latitude.

The search covers periods of 2–1000 h on a grid with step $0.5/T$, where T is the time span; the 30 deepest minima are then refined. A five times finer grid changed precision by less than one percentage point. When the first harmonic of the fit exceeds 1.20 times the second, the doubled period is adopted. On TUNE the PAB correction raises the fraction of correct periods from 44.2% to 52.2% and, at probability ≥ 0.8, the number of periods from 928 to 1389 at the same 92% precision.

4. Calibration and null tests

The truth set is LCDB V4.0 (Warner et al., 2009, 2021) with quality $U \geq 2-$, released in September 2021, before Gaia DR3. A period counts as correct if it is within 3% of the LCDB value. The 9,613 known asteroids were split by a hash of their number into TUNE (4,779) and TEST (4,834). LCDB contains errors, so precision measured against it is a lower bound.

Table 1: Signal-free asteroids (out of 32,229) that reach a given probability in the null tests.

Threshold	White	Block
0.70	264	248
0.75	104	108
0.80	39	32
0.85	12	8

On TUNE I fitted the doubling rule and a logistic model of the probability that a period is correct. Its seven features are the χ^2 gain over the no-rotation fit, the separation from the next minimum that is not an alias of the best one, the number of transits, the amplitude, the reduced χ^2, the period and the ratio of the two harmonics. On TEST, the predicted probabilities follow the observed fractions (Figure 1), except that the top bin (0.85–1) is overconfident: 0.95 predicted against 0.89 observed. The probability is therefore best read as a ranking with an approximate scale.

The confidence threshold was not chosen on TEST. Two null tests use the real Gaia sampling but remove the rotation. In the *white* test the residuals of the no-rotation model are permuted within seasons; in the *block* test one-day blocks of residuals are permuted, which keeps correlated systematics. Both were run on all 32,229 solved asteroids without an LCDB period. A noise period matches no catalogue, so every noise period that passes the threshold ends up among the candidate new periods. The rule is that the expected noise must stay below 5% of that list, with the probability scanned in steps of 0.05.

At the TUNE threshold for 80% precision (0.384), 20.0% (white) and 18.1% (block) of the signal-free asteroids pass (Table 1, Figure 2). The rule gives a threshold of 0.80, where 39 (white) and 32 (block) pass. Against the final list of 790 candidates (Section 5.6) that is 4.9% and 4.1%, with 95% Poisson upper limits of 51 and 43, or 6.5% and 5.4%. Permuted residuals may still hold some real signal, so these numbers are estimates rather than strict bounds.

Tier A is probability ≥ 0.80 and period < 100 h. Tier B is probability 0.384–0.80; it is not reliable for unknown asteroids, because 18–20% of signal-free asteroids reach it. Periods of 100–1000 h are flagged as slow candidates and never treated as confident.

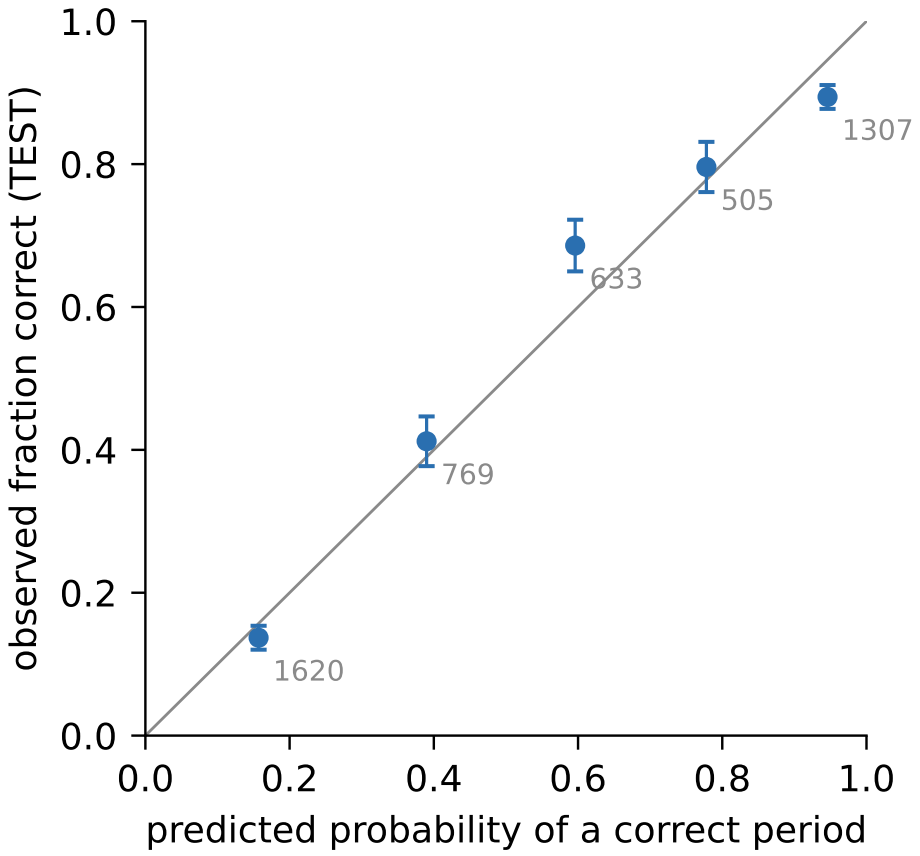


Figure 1: Reliability diagram on TEST for all solutions, in probability bins with edges 0, 0.3, 0.5, 0.7, 0.85 and 1. Bars are 95% binomial intervals; numbers give the count per bin.

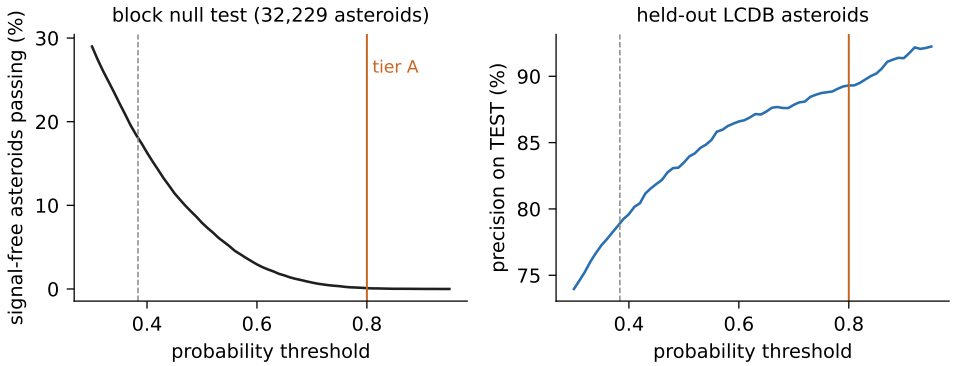


Figure 2: Left: fraction of signal-free asteroids (block null test) that pass, as a function of the probability threshold. Right: precision on TEST. Dashed line: the TUNE threshold for 80% precision. Solid line: the tier-A threshold.

5. Results

5.1. Periods

6,238 periods reach tier A. On TEST, 1303 of 1459 tier-A periods are correct: 89.3% (95% interval 87.6–90.8%); 90%, 88% and 83% at 2–10, 10–30 and 30–100 h (Figure 3). With tighter tolerances the figure is 87.9% within 1% and 82.7% within 0.3%. 1,680 tier-A periods are absent from all three earlier DR3 analyses (Ďurech & Hanuš, 2023; Cellino et al., 2024; MacLennan et al., 2026). It depends on the quality of the reference: 354 of 371 (95.4%, 92.8–97.1%) for $U \geq 3-$, 935 of 1049 (89.1%) for $U = 2$ and 2+, and 14 of 39 for $U = 2-$; without $U = 2-$ it is 90.8%.

Of the 156 disagreements, 58 differ from LCDB by exactly one cycle per day in lightcurve frequency, the typical alias of single-site ground photometry, 6 differ by a factor of two and 92 are other periods. For 38 of them an independent TESS period exists (Vavilov & Carry, 2025); it agrees with this work in 20 cases, with LCDB in 3, and with neither in 15.

Table 2 compares tier A with published catalogues. The three Gaia DR3 analyses share the data and agree at 99–99.6%; this tests consistency, not accuracy. Among independent surveys, the TESS periods of Vavilov & Carry (2025) agree for 91%. The ATLAS periods of Ďurech et al. (2022), which are not in VizieR and are therefore not a row of the table, agree for 1,088 of 1,176 (92.5%, 90.9–93.9%), and for 551 of 591 on TEST (93.2%). In the

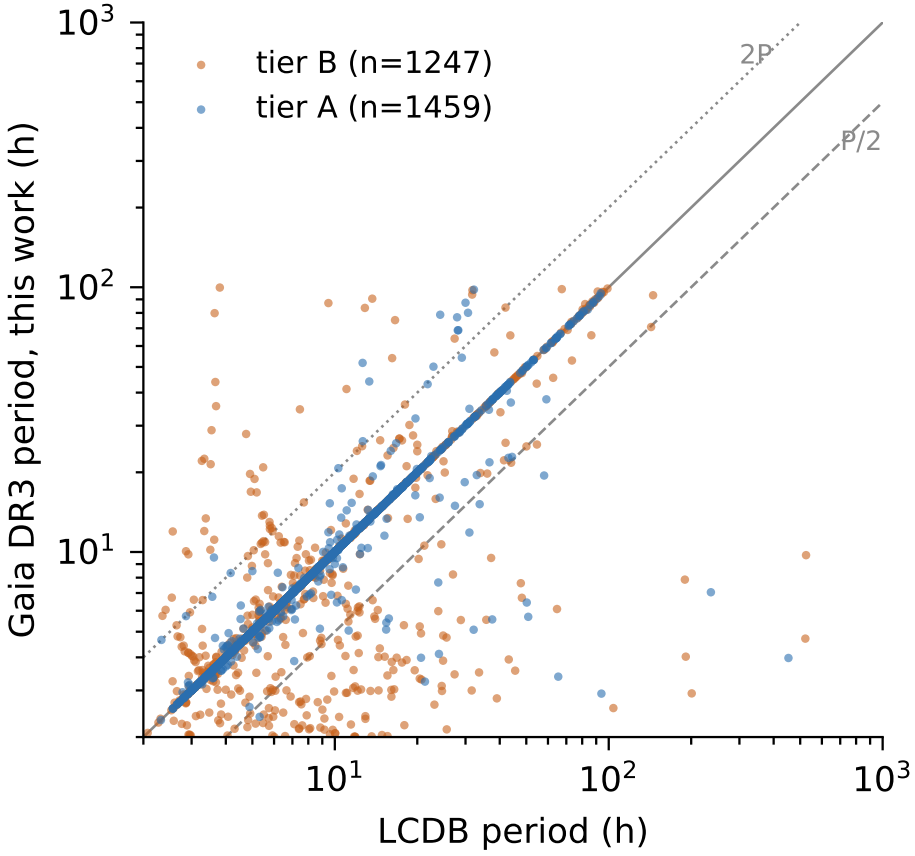


Figure 3: Period from this work against the LCDB period for TEST asteroids. Blue: tier A. Orange: tier B (probability 0.384–0.80). Lines mark P, $2P$ and $P/2$. The off-diagonal scatter includes daily aliases in the reference periods.

Table 2: Agreement of tier-A periods with published catalogues (within 3%).

Catalogue	Data	n	Agree
Ďurech & Hanuš (2023)	Gaia DR3	3564	99.6%
Cellino et al. (2024)	Gaia DR3	2883	99.0%
MacLennan et al. (2026)	Gaia DR3	2676	99.0%
Vavilov & Carry (2025)	TESS	759	91.0%
Erasmus et al. (2020)	ATLAS	280	68.6%
McNeill et al. (2023)	TESS	863	55.0%

Agreement measures consistency, not which side is right. It is 81% for Erasmus et al. entries with confidence $\geq 80\%$ ($n = 97$) and 77% for McNeill et al. entries with $CV_{\text{exact}} \geq 0.9$ ($n = 419$). Factor-two relations make up $\leq 1.2\%$ in every row. For TEST asteroids also in LCDB, agreement with LCDB (this work vs. the catalogue): 373 vs. 359 of 411 (Vavilov & Carry, 2025), 111 vs. 116 of 146 (Erasmus et al., 2020), and 352 vs. 231 of 379 (McNeill et al., 2023).

main disagreement, 72 periods twice theirs, neither side is shown to be right; for the 36 with an LCDB period, LCDB agrees with this work 31 times. The lower agreement with Erasmus et al. (2020) and McNeill et al. (2023) reflects those catalogues as much as this one: it rises to 81% and 77% for their own most confident entries, and against LCDB the periods of this work agree more often than theirs (table note).

The χ^2-curvature errors are too small. Against Ďurech & Hanuš (2023) the median $|\Delta P/P|$ is 3.9×10^{-5}; the catalogue adds $5.5 \times 10^{-5} P$ in quadrature, a value chosen so that the median $|\Delta P|/\sigma$ is 0.67, as for a normal distribution. This compares two analyses of the same data and does not cover aliasing.

5.2. Injection–recovery

I injected synthetic double-peaked lightcurves into the real Gaia sampling of 4,000 asteroids without a known period, with log-uniform periods of 2–100 h, amplitudes of 0.1–0.6 mag scaled by 0.5–1 per season, and a random sense with the phase following λ_{PAB}. The noise was either the object's own permuted residuals (pessimistic) or Gaussian at the quoted errors (optimistic). Tier A is precise under both: 96.5% (109/113) and 99.3% (567/571) of its periods are correct, with no $P/2$ or $2P$ errors, and the sense is correct for 96–97% of the correct periods. Because the injected signals have the

model's own form and an exact PAB shift, these figures are upper limits; the TEST and survey comparisons are the realistic ones. Completeness is low, 2.7% and 14%, against 10.5% of real unknown asteroids that reach tier A, and no amplitude below 0.2 mag is recovered. Tier A is a high-precision, low-completeness selection and should not be used for population statistics without a selection-function correction.

5.3. Spin senses

The sense s was compared with the hemisphere of published poles, excluding poles within 20° of the ecliptic and pole pairs in opposite hemispheres. It agrees for 3473 of 3535 asteroids (98.2%, 97.8–98.6%) with Ďurech & Hanuš (2023), which uses the same data; the 62 errors are nearly symmetric, 34 prograde called retrograde and 28 the reverse. It agrees for 3802 of 3899 (97.5%) with DAMIT (Ďurech et al., 2010), 372 of 394 (94.4%) with DAMIT asteroids not in Ďurech & Hanuš (2023), and 374 of 388 (96.4%, 94.0–97.8%) with the poles that Ďurech et al. (2020) derived from ATLAS photometry alone, 130 of 137 for those not in Ďurech & Hanuš (2023).

Accuracy depends on pole latitude, as expected from Section 3. Against Ďurech & Hanuš (2023), with the lower $|\beta|$ of the two pole solutions, it is 98.7% (3323/3366) for $|\beta| \geq 40°$, 88.8% (150/169) for $20° \leq |\beta| < 40°$, and 13 of 27 for the excluded poles below 20°. Tier A favours poles far from the ecliptic: among asteroids with poles in Ďurech & Hanuš (2023), $|\beta| < 30°$ for 1.7% of tier-A members and 15.5% of the others, probably because such poles give large amplitudes in every season. Most tier-A senses therefore belong to high-$|\beta|$ poles, but the accuracy of an individual sense with an unknown pole cannot be guaranteed.

1,447 tier-A asteroids have no pole in Ďurech & Hanuš (2023), DAMIT, Cellino et al. (2024), MacLennan et al. (2026), Ďurech et al. (2020) or Ďurech et al. (2016); for them the sense of rotation is new. The SsODNet spin compilation was not searched.

Tier A has 3,677 retrograde and 2,561 prograde rotators (59% retrograde). This excess is mostly a selection effect: applying the tier-A selection to the poles of Ďurech & Hanuš (2023) gives 54% retrograde among tier-A members against 40% among the others.

5.4. Yarkovsky segregation in families

The Yarkovsky drift moves prograde rotators outward and retrograde ones inward, so the two senses lie on opposite sides of a family's V-shape

(Vokrouhlický et al., 2015). This has been shown with modelled poles (Hanuš et al., 2013, 2018; Kryszczyńska, 2013) and, from Gaia DR3, by Ďurech & Hanuš (2023) in Eos, Eunomia, Themis and Koronis. Here it serves as a test of the senses and is quantified with a permutation null.

I used AstDyS family membership and synthetic proper elements (Knežević & Milani, 2003; Milani et al., 2014), with the proper a of the namesake as each family's centre a_c. A member is consistent if it is prograde and outside the centre, or retrograde and inside it. The null hypothesis keeps each family's mix of senses and its positions and breaks only the link between them, so I permuted the senses within each family 20,000 times. 1,753 of 2,345 tier-A members (74.8%) are consistent, against 52.4 ± 0.9% under the null ($z = 24.7$; no permutation reaches the observed value). In the 2×2 table of side (outside, inside) against sense (prograde, retrograde) the odds ratio is 9.1. Without Eos, which dominates, the fraction is 71.1%. With $d = |a - a_c|\, 10^{-0.2H}$, a distance in units of the V-shape that removes the $1/D$ dependence of the drift, normalised by each family's median, the fraction rises from 58% ($d < 0.5$) to 75% (0.5–1), 82% (1–2) and 82% ($d \geq 2$). With the median a of all members as centre it is 71.3%; for the 912 members without a pole in Ďurech & Hanuš (2023) it is 73.6%. The largest families show it clearly: (221) Eos 86% of 578, (24) Themis 81% of 184, (15) Eunomia 79% of 294, (158) Koronis 78% of 184 (Figure 4). Two smaller families, (93) Minerva and (375) Ursula, show the opposite pattern, which may reflect a namesake far from the family centre or interlopers.

For the 1,530 members with published poles, the senses of this work are correct for 98.2%, almost equally on the two sides (98.6% and 97.0%), and give a consistent fraction of 75.4% against 75.8% with the published poles. The 74.8% therefore measures the segregation itself, for a subset that favours high amplitudes and high $|\beta|$, where the drift is fastest; it does not describe complete families.

5.5. Disputed LCDB periods

I selected tier-A asteroids whose LCDB V4.0 period disagrees with this work and where either an independent survey supports this work and none supports LCDB (Vavilov & Carry, 2025; McNeill et al., 2023; Erasmus et al., 2020), or at least two other Gaia DR3 analyses support this work. This gives 117 asteroids, 32 of them exact daily aliases and 14 differing by a factor of two. In the 2023 LCDB revision distributed by the JPL Small-Body Database, 9

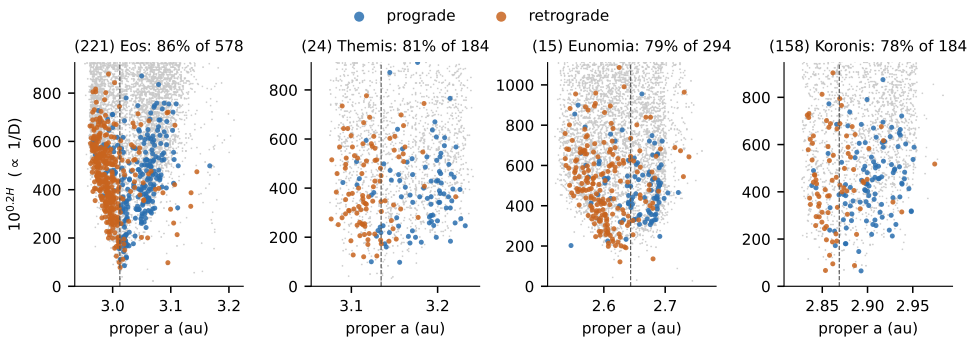


Figure 4: Four families in proper semimajor axis against $10^{0.2H} \propto 1/D$. Grey: all AstDyS members. Blue: prograde; orange: retrograde (tier A, this work). Dashed line: the namesake. Each title gives the fraction of members on the side the Yarkovsky effect predicts.

of the 117 had already been revised by others to the value found here, an independent sign that the selection finds real errors.

The other 108 are still in the LCDB, in two groups of different weight. The 39 in Table 3 disagree with independent survey data: a TESS or ATLAS period agrees with this work and none agrees with LCDB. They are not proven errors. The supporting surveys are less accurate than LCDB on average (Table 2), only 17 of the 39 are supported by two surveys, and 13 are factor-two cases, where sparse surveys share the same ambiguity. The TESS comparison above suggests that roughly half of such disagreements are LCDB errors. The six with $U \geq 3-$ deserve a closer look first. The other 69 are only *flagged*: they rest on Gaia DR3 analyses, which share the sampling and its aliases, and call for re-observation rather than correction. The full list of 117, with the supporting catalogues, is part of the data release.

5.6. Candidate new periods

942 tier-A asteroids have no LCDB period with $U \geq 2-$ and no period in the six catalogues of Table 2. I searched further: DAMIT; about 180 VizieR tables with rotation periods (5.0 M rows), with units from the table metadata and non-rotation periods excluded; the Minor Planet Bulletin, volumes 50–53; the ATLAS periods of Ďurech et al. (2022); the current JPL SBDB, which carries the October 2023 LCDB revision; and the SsODNet best-estimate table ssoBFT (Berthier et al., 2023), in a snapshot of 22 September 2026.

Table 3: The 39 LCDB V4.0 periods that disagree with independent survey data. Relation: d, daily alias; 2, factor of two; o, other. Survey periods: V, Vavilov & Carry (2025) (TESS); M, McNeill et al. (2023) (TESS); E, Erasmus et al. (2020) (ATLAS).

Number	$P_{\rm LCDB}$ (h)	U	P this work (h)	Rel.	Survey periods (h)
1040	59.2	2	37.7324	o	$37.736^{\rm V}$, $37.679^{\rm M}$
1940	6.953	3	6.5954	o	$6.596^{\rm V}$, $6.597^{\rm M}$
2884	14.31	2+	35.5264	o	$35.522^{\rm E}$
2933	13.482	2	13.0343	o	$13.033^{\rm V}$, $13.032^{\rm M}$
3422	2.91	3	3.2183	o	$3.218^{\rm V}$, $3.218^{\rm M}$
3813	12.3	3-	11.7155	o	$11.713^{\rm E}$
3867	28.035	2	51.0687	o	$51.031^{\rm V}$, $51.081^{\rm M}$
4488	24.25	2	78.7370	o	$78.640^{\rm E}$
5702	9.5	2+	10.3314	o	$10.333^{\rm E}$
8397	3.9	3-	4.4223	o	$4.422^{\rm V}$, $4.422^{\rm M}$
8744	5.199	2	5.3677	o	$5.367^{\rm E}$
8847	5.75	2	5.2301	o	$5.229^{\rm E}$
10338	10.55	2	13.4825	d	$13.482^{\rm E}$
10816	14.12	2	28.4494	2	$28.461^{\rm E}$
11087	4.5369	3	6.2802	o	$6.281^{\rm V}$
13894	5.9	2	5.4779	o	$5.477^{\rm V}$, $5.475^{\rm M}$, $5.478^{\rm E}$
14376	5.614	3	5.8485	o	$5.849^{\rm V}$, $5.848^{\rm M}$
17079	4.18	2+	3.8586	d	$3.859^{\rm E}$
17390	3.47535	2	6.9528	2	$6.948^{\rm V}$
17627	42.462	2	40.6427	o	$40.619^{\rm E}$
20623	19.936	2	20.5829	o	$20.562^{\rm V}$
23795	9.473	2	11.8136	d	$11.823^{\rm V}$, $11.810^{\rm M}$
26216	23.9952	2	4.1208	o	$4.111^{\rm V}$
28754	7.0243	2	14.0457	2	$14.048^{\rm V}$, $14.070^{\rm M}$
32358	21.82	2	43.0516	2	$43.252^{\rm V}$
34131	2.32495	2	4.6472	2	$4.651^{\rm V}$
37534	5.293	2	5.0942	o	$5.096^{\rm V}$
43089	36.2149	2	5.2126	o	$5.208^{\rm V}$
45659	3.848	2	3.5608	d	$3.557^{\rm V}$, $3.561^{\rm M}$
48694	94.0207	2	2.9181	o	$2.916^{\rm V}$
53324	3.59911	2	7.2018	2	$7.184^{\rm V}$
53396	1.41998	2	2.8409	2	$2.840^{\rm V}$
55098	5.71996	2	11.4198	2	$11.360^{\rm V}$
55543	2.86191	2	5.7249	2	$5.722^{\rm V}$, $5.725^{\rm M}$
57674	5.63622	2	11.2749	2	$11.266^{\rm V}$, $11.278^{\rm M}$
65441	3.55382	2	7.1060	2	$7.106^{\rm V}$, $7.107^{\rm M}$
91070	4.16574	2	8.3317	2	$8.331^{\rm V}$, $8.333^{\rm M}$
91256	1.43726	2	2.8745	2	$2.875^{\rm V}$, $2.874^{\rm E}$
109214	6.8107	2	5.7524	o	$5.749^{\rm V}$, $5.761^{\rm M}$

The snapshot contains 92% of the Ďurech et al. (2022) periods for asteroids without an LCDB period, so a few published periods may still be missed. Behrend's Geneva pages and ZTF period catalogues outside VizieR were not searched.

152 of the 942 (16.1%) have a period in one of these sources, including five whose LCDB entry has $U = 1$ or only a lower limit. For the 66 of them with a prior period from the last three sources and $U \neq 1$, this work agrees in 48 (73%, 61–82%), against 93.9% for the rest of tier A compared with the same sources (Ďurech et al. (2022), the current JPL SBDB and the SsODNet snapshot); against Ďurech et al. (2022) alone the figures are 35 of 46 (76%) and 93.2%. Most disagreements are simple relations (7 periods twice the prior one, 6 daily aliases, 2 at 3/2). Periods from the same Gaia data (Ďurech & Hanuš, 2023; Cellino et al., 2024; MacLennan et al., 2026) count against novelty but never as confirmation.

The remaining 790 are candidate new periods: 617 at 2–10 h, 116 at 10–30 h and 57 at 30–100 h. Two estimates bracket their reliability. TEST precision reweighted to the candidates' distribution of probability, amplitude and number of transits is 87%. Agreement with prior periods for the 66 comparable asteroids is 73%. The two differ, probably because asteroids without published periods are harder cases than the covariates capture. Between about 61%, the lower 95% bound of the agreement rate, and 87% of the 790 periods are therefore expected to be correct; noise accounts for 4–5%, and most of the rest are likely harmonics or aliases of the true period. For each candidate the catalogue gives the second-best period. The list is a set of targets for dense-lightcurve observers, biased to large amplitudes, high $|\beta|$ and periods above 2 h; it is not a population sample. An earlier release of the catalogue listed 861 candidates before the last three sources were searched.

Of the 42 candidates of the pilot (Tropynin, 2026a), 34 remain new; 4 are probably twice the pilot period, 3 now prefer slow solutions, and one, 11857, has a matching period in Ďurech et al. (2022).

6. Limitations and plans for Gaia DR4

Periods below 2 h are not searched, and tumblers fall outside a single-period model. The doubling rule based on the ratio of the two harmonics is a heuristic. When the Fourier fit is at half the rotation period, the PAB term should enter with a factor 2; a variant that also tries this changed TEST precision only from 89.3% to 89.7% and was not adopted. The probability

is slightly overconfident in its top bin. Completeness is low below 0.3 mag and nil below 0.2 mag, the injection test does not model aspect-dependent amplitudes, and spin senses are reliable mainly for poles far from the ecliptic.

DR4 will cover about 66 months. For it, I plan to fit the sidereal period and the pole jointly on a coarse pole grid, with a model-based test of P against $2P$, injections from DAMIT shape models with realistic poles, and a recalibrated probability. Thresholds and null tests can then be reused unchanged, so that results can be released within a day.

7. Conclusions

From Gaia DR3 sparse photometry alone, 6,238 asteroid rotation periods reach a tier whose precision is 89% against held-out LCDB periods and 92–93% against independent ATLAS periods. Each comes with a probability and a spin sense; the senses agree with independent ATLAS poles for 96% and reproduce the Yarkovsky segregation in families. For asteroids without a published period, the error rate is higher than for known ones, and this paper measures by how much: 790 candidate new periods carry an estimated 4–5% noise, and a further 8–35% are probably harmonics or aliases. Observers can use the candidate list and the 39 LCDB periods that disagree with survey data as target lists; modellers can use the 1,447 new senses with the latitude caveat. The same pipeline, thresholds and null tests are ready for Gaia DR4.

CRediT authorship contribution statement

Oleg Tropynin: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) to draft and edit the text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication. Figures were plotted with Matplotlib from the released data; no generative AI was used to create images.

Data availability

The catalogue (Tropynin, 2026b) (41,842 rows with periods, probabilities, tiers, second-best periods and spin senses), the list of 790 candidate new periods with the prior periods found for the other 152, the 117 disputed LCDB periods with their supporting catalogues, the validation tables and the full code are openly available on Zenodo, doi:10.5281/zenodo.23638281, and at `https://github.com/olegtropinin/gaia-dr3-rotation-catalog`.

Acknowledgements

This work has made use of data from the European Space Agency (ESA) mission Gaia (`https://www.cosmos.esa.int/gaia`), processed by the Gaia Data Processing and Analysis Consortium (DPAC, `https://www.cosmos.esa.int/web/gaia/dpac/consortium`). Funding for the DPAC has been provided by national institutions, in particular the institutions participating in the Gaia Multilateral Agreement. It also used the JPL Horizons system and Small-Body Database, the LCDB, DAMIT, AstDyS, SsODNet (IMCCE) and VizieR (CDS, Strasbourg).

References

Berthier, J., Carry, B., Mahlke, M., & Normand, J. 2023, A&A, 671, A151

Cellino, A., et al. 2024, A&A, 687, A277

Ďurech, J., & Hanuš, J. 2023, A&A, 675, A24

Ďurech, J., Hanuš, J., Oszkiewicz, D., & Vančo, R. 2016, A&A, 587, A48

Ďurech, J., Tonry, J., Erasmus, N., et al. 2020, A&A, 643, A59

Ďurech, J., Vávra, M., Vančo, R., & Erasmus, N. 2022, Front. Astron. Space Sci., 9, 809771

Ďurech, J., Sidorin, V., & Kaasalainen, M. 2010, A&A, 513, A46

Erasmus, N., et al. 2020, ApJS, 247, 13

Gaia Collaboration, Prusti, T., et al. 2016, A&A, 595, A1

Gaia Collaboration, Vallenari, A., et al. 2023a, A&A, 674, A1

Gaia Collaboration, Tanga, P., et al. 2023b, A&A, 674, A12

Hanuš, J., et al. 2013, A&A, 559, A134

Hanuš, J., et al. 2018, Icarus, 299, 84

Harris, A. W., Young, J. W., Scaltriti, F., & Zappalà, V. 1984, Icarus, 57, 251

Kaasalainen, M. 2004, A&A, 422, L39

Knežević, Z., & Milani, A. 2003, A&A, 403, 1165

Kryszczyńska, A. 2013, A&A, 551, A102

MacLennan, E. M., et al. 2026, A&A, 707, A131

McNeill, A., et al. 2023, AJ, 166, 152

Milani, A., Cellino, A., Knežević, Z., et al. 2014, Icarus, 239, 46

Tropynin, O. 2026a, Candidate rotation periods for main-belt asteroids from Gaia DR3 sparse photometry, Zenodo, doi:`https://doi.org/10.5281/zenodo.23624115`

Tropynin, O. 2026b, Rotation periods and spin senses of asteroids from Gaia DR3 sparse photometry, Zenodo, doi:`https://doi.org/10.5281/zenodo.23638281`

Vavilov, D., & Carry, B. 2025, A&A, 693, A66

Vokrouhlický, D., Bottke, W. F., Chesley, S. R., Scheeres, D. J., & Statler, T. S. 2015, in Asteroids IV, ed. P. Michel, F. E. DeMeo, & W. F. Bottke (Tucson: Univ. Arizona Press), 509

Warner, B. D., Harris, A. W., & Pravec, P. 2009, Icarus, 202, 134

Warner, B. D., Harris, A. W., & Pravec, P. 2021, Asteroid Lightcurve Database (LCDB) Bundle V4.0, NASA Planetary Data System, doi:10.26033/j3xc-3359